Statistics for Data Science -1

Tutorial: Permutations and combinations and Probability

March 17, 2024

Basic principles of counting

Permutations

Combinations

Applications: Permutations or combinations

Probability

Random Experiment, Sample Space, Events

Properties of Probability

Conditional Probability and Bayes Theorem

Probability trees

Addition rule of counting

Shirt 3

- If an action A can occur in n_1 different ways, another action B can occur in n_2 different ways, then the total number of occurrence of the actions A **or** B is $n_1 + n_2$.

Pant 2

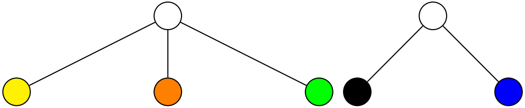
Shirt or Pant 3+2

Multiplication rule of counting

- If an action A can occur in n_1 different ways, another action B can occur in n_2 different ways, then the total number of occurrence of the actions A **and** B together is $n_1 \times n_2$. $3 \times 2 = 6$
- Suppose that r actions are to be performed in a definite order. Further suppose that there are n_1 possibilities for the first action and that corresponding to each of these possibilities are n_2 possibilities for the second action, and so on. Then there are $n_1 \times n_2 \times \ldots \times n_r$ possibilities altogether for the r actions.

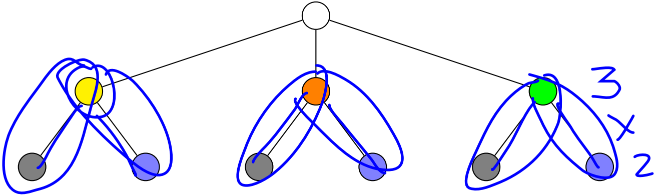
Illustration

- Choose a shirt or choose a pant



Addition rule: $3 + 2 = 5$ ways

- Choose a shirt and a pant



Multiplication rule: $3 \times 2 = 6$ ways

Factorial

$n_1 \times n_2 \times n_3$

Definition

The product of the first n positive integers (counting numbers) is called n factorial and is denoted n!. In symbols,

$$n! = n \times (n-1) \times \ldots \times 1$$

$n \times n-1 \times n-2 \times \ldots 1$

Example:

$$5 \to 4 \to 3 \to 2 \to 1$$

$$n = 5$$

In this example, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Remark

By convention $0! = 1$

Permutation

A B C $n = 3$

1 A B
2 A C
3 B A
4 B C
5 C A
6 C B

$^3P_2 = 3!$

AB
AC
BC

Definition

A *permutation* *is an ordered arrangement of all or some of n objects.*

$n = 3$ $r = 2$ $^nP_r = {}^3P_2 = 6$

$n \times (n-1)$

$n \times n \times n \times \quad \times n$

Permutation when objects are distinct- repetitions not allowed

The number of possible permutations of r objects from a collection of n distinct objects is given by the formula

$$n \times (n-1) \times \ldots \times (n-r+1)$$

and is denoted by ^nP_r

$$^nP_r = \frac{n!}{(n-r)!} \rightarrow$$

$r!$

$n \times n-1 \times n-2 \quad \times n-r+1$

$r!$

$5 \rightarrow 3$

- Special cases
  1. $^nP_0 = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$ There is only one ordered arrangement of 0 objects.
  2. $^nP_1 = \frac{n!}{(n-1)!} = n$. There are n ways of choosing one object from n objects.
  3. $^nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$. We can ararnge n distinct objects in $n!$ ways- multiplication principle of counting.

Permutation when objects are distinct- repetitions allowed

The number of possible permutations of r objects from a collection of n distinct objects when repetition is allowed is given by the formula

$$n \times n \times \ldots \times n$$

and is denoted by n^r

$\boxed{n}$ Rep alw $\quad {}^nP_r = \dfrac{n!}{(n-r)!}$

Rep X $\quad n^r$

Permutation when objects are not distinct

- The number of permutations of n objects when p of them are of one kind and rest distinct is equal to

$$\frac{n!}{p!}$$

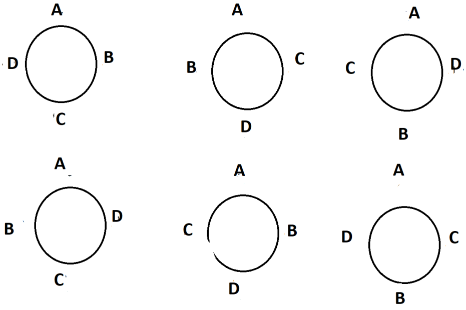
- The number of permutations of n objects where p_1 is of one kind, p_2 is of second kind, and so on p_k of k^{th} kind is given by

$$\frac{n!}{p_1!p_2!\dots p_k!}$$

- Applying the above formula to the word " STATISTICS"; $n = 10, p_1 = 3, p_2 = 3, p_3 = 1, p_4 = 2, p_5 = 1$. Hence, total number of ways =

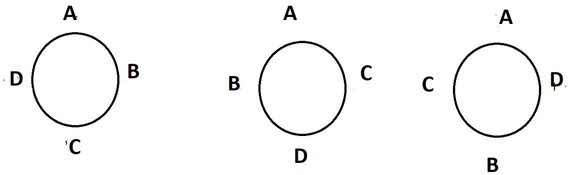
$$\frac{10!}{3!3!1!2!1!} = 50,400$$

Circular permutation: Clockwise and anticlockwise are different



The number of ways n distinct objects can be arranged in a circle (clockwise and anticlockwise are different) is equal to $(n-1)!$

Circular permutation: Clockwise and anticlockwise are same



The number of ways n distinct objects can be arranged in a circle (clockwise and anticlockwise are same) is equal to $\frac{(n-1)!}{2}$

Combinations: Notation and formula

- In general, each combination of r objects from n objects can give rise to $r!$ arrangements.
- The number of possible combinations of r objects from a collection of n distinct objects is is denoted by ${}^nC_r$ and is given by

$$^nC_r = \frac{n!}{r!(n-r)!}$$

$$\frac{n!}{r!(n-r)!} \qquad \frac{^nP_r}{r!}$$

- Another common notation is $\binom{n}{r}$ which is also referred to as the binomial coefficient

$$^3P_2 = 6 = {}^3C_2 \cdot 2!$$

Some useful results

1. ${}^{n}C_{r} = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)!r!} = {}^{n}C_{(n-r)}$
   In other words, selecting r objects from n objects is the same as rejecting $n-r$ objects from n objects.
2. ${}^{n}C_{n} = 1$ and ${}^{n}C_{0} = 1$ for all values of n
3. ${}^{n}C_{r} = {}^{n-1}C_{r-1} + {}^{n-1}C_{r}; 1 \leq r \leq n$

Applications: Permutations or combinations

- Important to distinguish between situations involving combinations and situations involving permutations.
- Permutation- "order matters". Combination - "order does not matter"

Permutation or combination

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Permutation or combination: Forming/choosing teams

- Forming teams: If the team members' roles or positions are distinct, and the order of selection matters (e.g., choosing a team leader, presenter, and coordinator), you'd use permutations. However, if the team members are treated equally, and the order of selection doesn't matter, you'd use combinations

  1. In how many ways can a committee of 4 members be selected from a group of 10 people ?
  2. In how many ways can a committee of 4 members be selected from a group of 10 people to hold specific positions: President, Vice President, Treasurer, and Secretary?

Forming a 4-letter code

- Forming a 4-letter code using the letters "A," "B," "C," and "D."- Since each letter can only be used once, and the order matters, we need to use permutations.
- Each letter can be used only once in the code, and the order of the letters matters.

Section summary

- Need to distinguish between permutation and combination.
- Examples of situations where permutation is applied, combination is applied.

Random experiment

$S = \{ RCB, DC \}$

Definition

An experiment is any process that produces an observation or outcome.

Definition

A random experiment is an experiment whose outcome is not predictable with certainty.

Remark

However, although the outcome of the experiment will not be known in advance, let us suppose that the set of all possible outcomes is known.

Sample Space

Definition

A sample space *(denoted by Ω or S) : collection of all basic outcomes.*

- Basic Outcomes: the possible outcomes that can occur must be:
  1. mutually exclusive: only one basic outcome can occur
  2. exhaustive: one basic outcome must occur

Events

Definition

An event E is a collection of basic outcomes.

- That is, an event is a subset of the sample space.
- We say an event has occurred if the outcome is contained in the subset.

Union of events

- For any two events E and F, we define the new event $E \cup F$ called the union of events E and F, to consist of all outcomes that are in E or in F or in both E and F.
- That is, the event $E \cup F$ will occur if either E or F occurs.

Intersection of events

- For any two events E and F, we define the new event $E \cap F$ called the intersection of events E and F, to consist of all outcomes that are in E and in F.
- That is, the event $E \cap F$ will occur if both E and F occurs.

Example: Tossing two coins

two exams

- Experiment: Tossing ~~two coins~~ and noting the outcomes
  Event:
  - Sample space: {PP, PF, FP, FF} {M_1M_2, M_1F_2, F_1M_2, F_1F_2}
  - head on the first toss {PP, PF}
  - head on second toss {PP, FP}
  - head on first or second toss {PP, PF, FP}
  - head on first and second toss {PP}

Example: Tossing two coins

- Experiment: Tossing two coins and noting the outcomes
  Event:
  - Sample space: $S = \{HH, HT, TH, TT\}$
  - head on the first toss $E_1 = \{HH, HT\}$
  - head on second toss $E_2 = \{HH, TH\}$
  - head on first or second toss $E_1 \cup E_2 = \{HH, HT, TH\}$
  - head on first and second toss $E_1 \cap E_2 = \{HH\}$

Null event and disjoint event

Definition

We call the event without any outcomes the null event, and designate it as ϕ

Definition

If the intersection of E *and* F *is the null event, then since* E *and* F *cannot simultaneously occur, we say that* E *and* F *are disjoint, or mutually exclusive.*

Complement of an event

Definition

The complement of E, denoted by E^c, consists of all outcomes in the sample space S that are not in E.

- That is, E^c will occur if and only if E does not occur.
- The complement of the sample space is the null set, that is $S^c = \Phi$

Examples of complement of an event

- Experiment: Toss a coin once and note the outcomes
  - Sample space: $S = \{H, T\}$
  - Event E_1: out come is head $E_1 = \{H\}$
  - Event E_2: out come is tail $E_2 = \{T\}$
  - Event E_2 is complement of event E_1. In other words, $E_2 = E_1^c$
- Experiment: Tossing two coins and noting the outcomes
  - Sample space: $S = \{HH, HT, TH, TT\}$
  - Event: head on the first toss $E_1 = \{HH, HT\}$
  - $E_1^c = \{TH, TT\}$; tail on first toss

Subsets

Definition

For any two events E and F, if all of the outcomes in E are also in F, then we say that E is contained in F, or E is a subset of F, and denote it as $E \subset F$

- Example: Experiment: Tossing two coins and noting the outcomes
  - Sample space: $S = \{HH, HT, TH, TT\}$
  - Event: head on the first toss $F = \{HH, HT\}$
  - Event: head in both the tosses $E = \{HH\}$
  - $E \subset F$

Section summary

1. Notion of events
2. Union, intersection, complement of events
3. Null event and mutually exclusive (disjoint) events

Venn diagrams

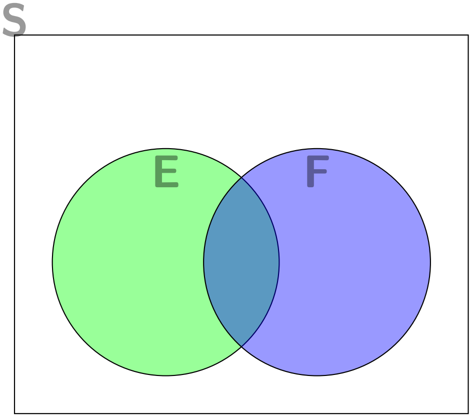
- A graphical representation that is useful for illustrating logical relations among events is the Venn diagram.

Representation of sample space

- Representation of sample space: Sample space consists of all possible outcomes and is represented by a large rectangle.

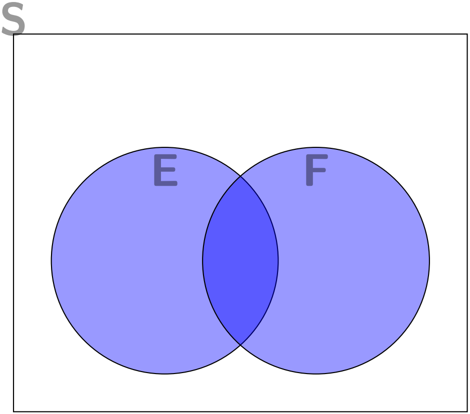
Representation of event

- Representation of event: The events $E, F, G, \ldots$ are represented in given circles within the rectangle.

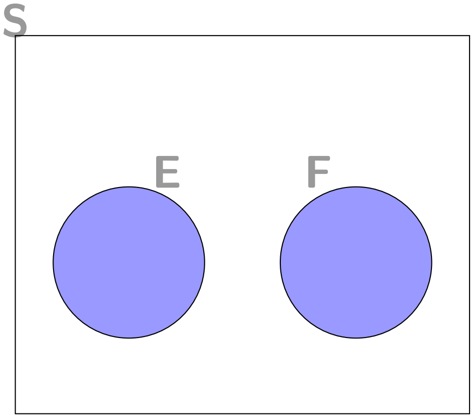


Representation of event: union and intersection

- Representation of event:
  - $E \cup F$ is entire shaded region
  - $E \cap F$ is the shaded in blue region

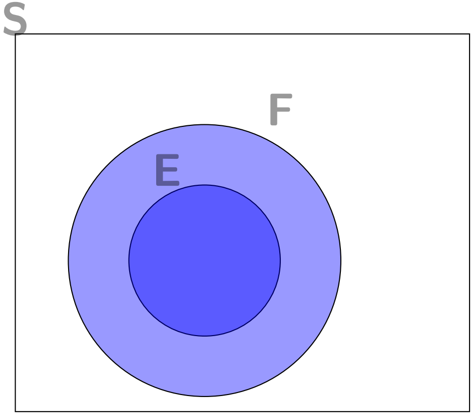


Representation of event: disjoint events



Representation of event: subsets

$E \subset F$



Topic summary

1. Introduced random experiment, sample space, event.
2. Notion of union, intersection, complement of events.
3. Representation of sample space, events, using venn diagrams.

Probability Axioms

Consider an experiment whose sample space is S. We suppose that for each event E there is a number, denoted $P(E)$ and called the probability of event E, that is in accord with the following three properties (axioms).

1. For any event E, the probability of E is a number between 0 and 1. That is, $0 \leq P(E) \leq 1$.
2. The probability of sample space S is 1. Symbolically, $P(S) = 1$. In other words, the outcome of the experiment will be an element of sample space S with probability 1.
3. For a sequence of mutually exclusive (disjoint) events, $E_1, E_2, \ldots,$

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Probability of union of disjoint events

The third property can be stated as:
The probability of the union of disjoint events is equal to the sum of the probabilities of these events.
For instance, if E_1 and E_2 are disjoint, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

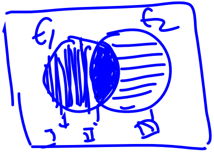
In other words, if events E_1 and E_2 cannot simultaneously occur, then the probability that the outcome of the experiment is contained in either E_1 or E_2 is equal to the sum of the probability that it is in E_1 and the probability that it is in E_2.

General properties of probability

Properties 1, 2, and 3 can be used to establish some general results concerning probabilities.

1. Probability of complement of an event: $P(E^c) = 1 - P(E)$
   - E and E^c are disjoint. Also, $E \cup E^c = S$
   - Apply Property 3 to LHS $P(E \cup E^c) = P(E) + P(E^c)$
   - Apply Property 2 to RHS $P(S) = 1$
   - Equating both, we get $P(E \cup E^c) = P(E) + P(E^c) = P(S) = 1$. Hence $P(E^c) = 1 - P(E)$
2. $P(\Phi) = 0$
   - $S^c = \Phi$
   - Apply the above property, $P(S^c) = 1 - P(S)$
   - Apply property 2, $P(S) = 1$
   - Hence $P(\Phi) = 0$

Addition rule of probability



The following formula relates the probability of the union of events E_1 and E_2, which are not necessarily disjoint, to $P(E_1), P(E_2)$, and the probability of the intersection of E_1 and E_2. It is often called the addition rule of probability.

For any events E_1 and E_2,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$P(E_1 \cup E_2) = P(\mathrm{I} \cup \mathrm{II} \cup \mathrm{III}) = P(\mathrm{I}) + P(\mathrm{II}) + P(\mathrm{III})$$

$$P(E_1) - P(E_1 \cap E_2) + P(E_1 \cap E_2) + P(E_2) - P(E_1 \cap E_2)$$

Example: Shopping for shirts and pants

A customer that goes to the clothng store will purchase a shirt with probability 0.3. The customer will purchase a pant with probability 0.2 and will purchase both a shirt and a pant with probability 0.1. What proportion of customers purchases neither a shirt nor a pant?

- Let S denote the event of a customer purchasing a shirt $P(S) = 0.3$
- Let P denote the event of a customer purchasing a pant $P(P) = 0.2$
- Proportion of customers purchases neither a shirt nor a pant?
- Neither a shirt nor a pant is complement of the event of either shirt or pant.

$$P(S \cup P) = P(S) + P(P) - P(S \cap P)$$
$$= 0.3 + 0.2 - 0.1$$
$$= 0.4$$

$$P(S \cup P)^C = 0.6$$

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- $P(S \cup P) = P(S) + P(P) - P(S \cap P) = 0.3 + 0.2 - 0.1 = 0.4$

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- We know $P(S \cup P)^c = 1 - P(S \cup P)$
- $P(S \cup P) = P(S) + P(P) - P(S \cap P) = 0.3 + 0.2 - 0.1 = 0.4$
- Hence, $P(S \cup P)^c = 1 - 0.4 = 0.6$

Equally likely outcomes

- For certain experiments it is natural to assume that each outcome in the sample space S is equally likely to occur.
- That is, if sample space S consists of N outcomes, say, $S = \{1, 2, \ldots, N\}$, then it is often reasonable to suppose that

$$P(\{1\}) = P(\{2\}) = \cdots = P(\{N\})$$

- In this expression, $P(\{i\})$ is the probability of the event consisting of the single outcome i.
- Using the properties of probability, we can show that the foregoing implies that the probability of any event A is equal to the proportion of the outcomes in the sample space that is in A.
- That is, $P(A) = \frac{\text{number of outcomes in } S \text{ that are in } A}{N}$

Example: Rolling a dice

- Experiment: Roll a fair dice
- Sample space: $S = \{1, 2, 3, 4, 5, 6\}$
- Let E_i denote the event of outcome i. Since the dice is fair, $P(E_i) = \frac{1}{6}$.
- Define A to be the event the outcome is odd $A = \{1, 3, 5\}$

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  $P(A) = P(E_1) + P(E_3) + P(E_5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$
- Let B be the event that the outcome is greater than 4.
  $B = \{5, 6\}$

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- Let B be the event that the outcome is greater than 4.
  $B = \{5, 6\}$ $P(B) = \frac{2}{6}$
- Let C be the event that the outcome is either odd or greater than 4.

KA

Example: Rolling a dice

K = 4/52 A(K)

A = 4/52

S: {c1 c2 52.52

- Experiment: Roll a fair dice
- Sample space: $S = \{1, 2, 3, 4, 5, 6\}$
- Let E_i denote the event of outcome i. Since the dice is fair, $P(E_i) = \frac{1}{6}$.
- Define A to be the event the outcome is odd $A = \{1, 3, 5\}$
  $P(A) = P(E_1) + P(E_3) + P(E_5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$
- Let B be the event that the outcome is greater than 4.
  $B = \{5, 6\}$ $P(B) = \frac{2}{6}$
- Let C be the event that the outcome is either odd or greater than 4.
  $P(C) = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6}$

Section summary

1. Interpretations of probability.
2. Probability axioms
3. Addition rule of probability.

Conditional Probability and Independent Events

- What is the probability that event E occurs given that event F occurs (or conditional on event F occurring)?

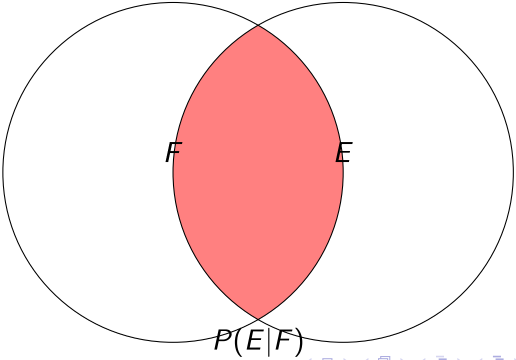
$S = \{HH, HT, TH, TT\}$

$E_1 = \{HH, HT\}$

$E_2 = \{HH, TH\}$

$E_3 :$

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$



Multiplication rule

- Conditional probability formula:

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

  Multiply both sides of the formula with $P(F)$, we get

- $$P(E \cap F) = P(F)P(E|F)$$

- This rule states that the probability that both E and F occur is equal to the probability that F occurs multiplied by the conditional probability of E given that F occurs.
- It is often quite useful for computing the probability of an intersection.

Generalized multiplication rule

A generalization of the multiplication rule, which provides an expression for the probability of the intersection of an arbitrary number of events, is referred to as the generalized multiplication rule and is given by

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$$P(E_1 \cap E_2 \cap \ldots \cap E_n) =$$
$$P(E_1)P(E_2|E_1)P(E_3|E_1 \cap E_2) \ldots P(E_n|E_1 \cap E_2 \ldots \cap E_{n-1})$$

Independent events: definition

Independent events: definition

Since

$$P(E \cap F) = P(F) \times P(E|F)$$

Independent events: definition

Since

$$P(E \cap F) = P(F) \times P(E|F)$$

we see that E is independent of F if

$$P(E \cap F) = P(F) \times P(E)$$

Definition

Two events E and F are independent if $P(E \cap F) = P(E) \times P(F)$.

Definition

Two events that are not independent are said to be dependent.

Bayes Theorem

Suppose A_1 and A_2 are mutually exclusive and exhaustive events. Consider $P(A_1|B)$. This probability can be written

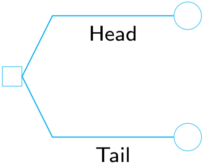
$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2)}$$

Exercise

1. Three manufacturing plants A, B, and C supply 20, 30 and 50%, respectively of all shock absorbers used by a certain automobile manufacturer. Records show that the percentage of defective items produced by A, B and C is 3, 2 and 1%, respectively.
   1.1 What is the probability that a randomly chosen shock absorber installed by the manufacturer will be defective?
   1.2 What is the probability that the part came from manufacturer A, given that the part was defective?

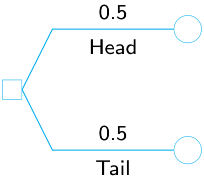
Experiment: Toss a coin once

$S = \{H, T\}$



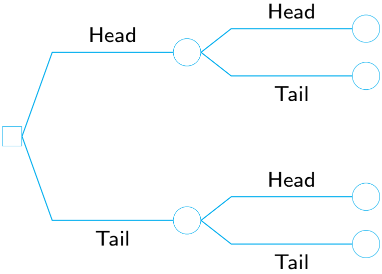
Experiment: Toss a coin once

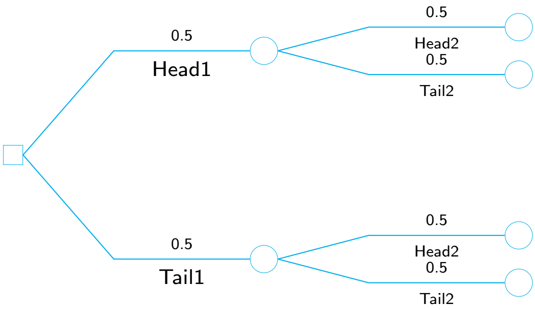
$S = \{H, T\}$ with $P\{H\} = P\{T\} = 0.5$



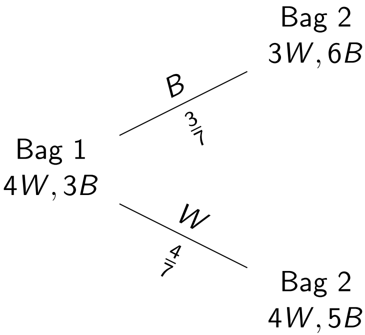
Experiment: Toss a coin twice

$$S = \{HH, HT, TH, TT\}$$

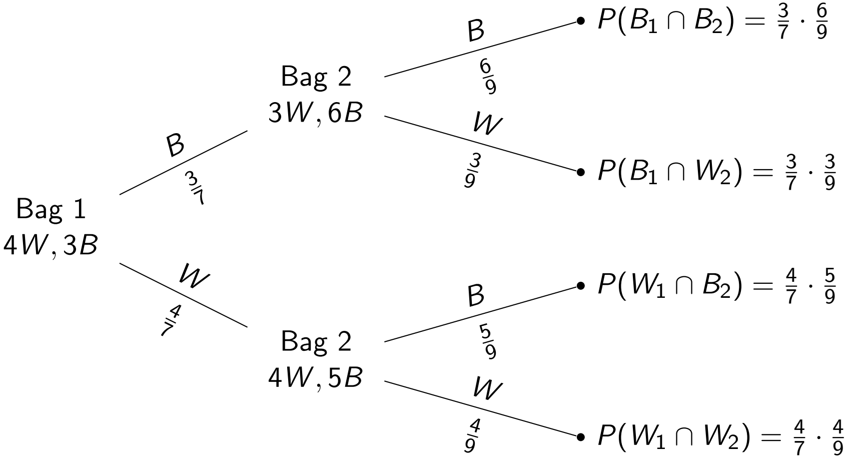



0.5
Head1
0.5
Tail1
0.5
Head2
0.5
Tail2
0.5
Head2
0.5
Tail2

Pick a ball from a bag of 3 Black and 4 White balls



Pick a ball from two bags



Thank you